Assignment 4 Part A & B Peer Evaluation

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Part A Figure

1. Looking at the <u>figure only</u>, write down what you think is "the goal" of the proposed program (This Proposal)?
    a. Based on the figure alone, it seems that the goal of this proposal is to utilize kinematic lensing results from Roman along with weak lensing in a 3x2 style analysis to better constrain the stellar mass-halo mass relation and reduce systematic effects/break degeneracies.

2. Looking at the <u>figure only</u>, write down what you think is "the target" of the proposed program ?
    a. From the figure alone, the target for this proposal appears to be galaxy weak lensing and kinematic lensing measurements from Roman.

3. Looking at the <u>figure only</u>, write down what you think "the problem" is that the proposed program is trying to address?
    a. From the figure, the problem this proposal is trying to solve is that constraints on the stellar mass-halo mass relation using weak lensing results alone has systematic effects that cannot be fully described without another tracer such as kinematic lensing.

4. Now look at goal statement in the caption – did what you think is "the goal" and "target" match the actual "goal" and "target" of the program?
    a. Figure Goal:
        i. to <u>constrain the dwarf-halo mass scale and thereby improve the reliability of low-mass SHMR constraints central to galaxy–halo connection and dark-matter tests.</u>
        ii. This is in-line with what I assumed from the figure alone, although the mass-scale systematics was not readily apparent as someone who is not well-versed in dark matter/galaxy studies.

b. Figure target:

   i. Roman Medium+Wide-tier stacked weak lensing, calibrated by the Deep tier and strengthened by WL×KL

   ii. This is also in-line with what I assumed from the figure alone, although the language includes jargon that I am not familiar with.

5. Now look at problem statement in the caption – did what you think "the problem" is match the actual "problem" posed in the abstract?

   a. Caption Problem:

      i. The key problem is that low-mass SHMR inferences can be limited by **mass-scale systematics** tied to modeling assumptions rather than by statistical errors alone, motivating an independent calibration.

   b. Abstract Problem:

      i. At the low-mass end, many SHMR constraints rely on indirect halo-mass inference, leaving systematic uncertainties from selection, baryonic physics, and environment largely uncalibrated by an independent mass probe.

   c. These match up well.

6. How can the figure be modified to better illustrate "the goal" and/or "the problem" of the program? Are all the components of the figure needed?

   a. Visually, it looks like the goal is to go from the green error bars to the red. As for the problem, it took some time to realize that the abundance matching line (orange) is the comparison, noting the larger deviations of the inferred SHMR Z&B fit at lower masses. I think every thing here is needed. The story could be clearer. Another red line and scatter region would be helpful in showing the systematic improvements in the fits in Halo vs Stellar mass space. Main limitation for understanding the figure is the jargon. Also, it's unclear if the goal is just for the 2x10^10 point or if its also for the other binned points.

Part A Figure Caption

1. Does the first bold face statement convey a convincing *statement* for a big picture for the program? Any suggestions for improvement?
    a. This is pretty good. It is a powerful statement and the impacts are important broadly. No improvements needed.

2. Do the next sentences sufficiently describe the plot? Any suggestions for improvement?
    a. Suggest adding (Z&B) after Zaritsky & Behroozi to emphasize the acronym in the legend. Do not know what HLWAS is; add explanation.

3. Is the problem statement clear? Any suggestions for improvement?
    a. This is very clear. Good; keep as is.

4. Is the goal statement clear and underlined? Any suggestions for improvement?
    a. Yes and yes. This is clear. Good, no suggestions for improvement.

5. Last sentence: Is the importance of the program convincing? Any suggestions for improvement?
    a. Could have a stronger wording. E.g. "improve the reliability" sounds weaker than something like just "improves the low-mass…"

Part B Abstract

1. Write down the problem that the talk is trying to address. (Does it start with "However"??)
    a. At the low-mass end, many SHMR constraints rely on indirect halo-mass inference, leaving systematic uncertainties from selection, baryonic physics, and environment largely uncalibrated by an independent mass probe.
    b. Does not start with "However".

2. Is this problem effectively conveyed in the title? How might the title be improved?

   a. No mention of low mass end of the SHMR in the title, which is the central topic.

3. Do the facts and importance (first 2 sentences) sufficiently set the stage for the problem statement? Any suggestions for improvement?
   a. These are a very good introduction to the topic and the importance. No changes here. Keep as is.

4. Is the goal statement ("In this talk") sufficiently clear? Any suggestions for improvement?
   a. This is a jargon thing but I do not know what stacking of lensing signals does and how that improves constraints. Adding a brief explanation might be helpful but dependent on audience.

5. Is the statement of urgency/importance of the talk topic convincing? Any suggestions for improvement?
   a. The importance is apparent. The urgency is not convincing; I do not understand "why now?". I think answering that question in this section could be helpful for understanding the urgency.

6. Last sentence – are the broader impacts of the talk topic convincing ? Any suggestions for improvement?
   a. This is good. Keep as is. No changes needed.